Graph-Constrained Reasoning for Reliable LLM Question Answering

Carlos Gonzalez (uc2025195081) João Lino (uc2025246120)

1. Problem and Goals

Large Language Models (LLMs) can hallucinate, use outdated information and make opaque decisions [3, 1]. Knowledge Graphs (KGs) provide structured knowledge to mitigate these issues [3], but efficiently constraining LLM reasoning to KG paths remains challenging [1, 2].

Research Question: How can KG-constrained LLM reasoning reduce unfaithful reasoning and improve factual accuracy in complex Question Answering tasks?

- **Goal 1:** Implement a KGQA pipeline combining LLM reasoning with KG path exploration.
- **Goal 2:** Evaluate answer accuracy and reasoning faithfulness on a benchmark such as WebQSP or CWQ.
- **Goal 3:** Compare the approach with a baseline, such as a zero-shot LLM without KG access, to assess reductions in unfaithful reasoning.

2. Related Work and Motivation

Our work builds on three recent methods. Plan-on-Graph (PoG) [3] lets an LLM adaptively explore and self-correct KG paths, but at the cost of many iterative LLM calls. G-Retriever [2] retrieves a compact, relevant subgraph via a Prize-Collecting Steiner Tree (PCST), but leaves the final generation unconstrained. Graph-constrained Reasoning (GCR) [1] constrains decoding to valid KG paths through a KG-Trie, but indexes all L-hop paths regardless of relevance, so the trie grows and adds noise as L increases.

Motivation: We propose a hybrid pipeline: G-Retriever's retrieval isolates a compact subgraph, on which we build a localized KG-Trie to apply GCR's decoding constraints. This combines GraphRAG's scalability with KG-grounded reasoning paths.

3. Data and Tools

The hybrid pipeline will be assessed on the multi-hop question-answering benchmarks **WebQSP** and **ComplexWebQuestions (CWQ)**, using **Freebase** as the underlying KG. For questions outside the benchmarks, the subgraph is obtained live from **Wikidata** via **SPARQL** queries around the linked question entities, and visualized (e.g., with NetworkX/PyVis).

Following recent research, a lightweight LM such as **SentenceBERT** will encode the graph attributes, an open-source LLM such as **Llama-3-8B** will be used for graph-constrained decoding, and a general-purpose LLM such as **GPT-4o-mini** will produce the final answer.

Tools and Hardware: PyTorch, Hugging Face Transformers and NetworkX, on local or cloud GPUs.

4. Approach

Our pipeline combines language and explicit KG knowledge in three stages:

1. **Graph Acquisition:** entities in the question are linked and their L-hop neighbourhood is retrieved from the KG (SPARQL for Wikidata; the dataset subgraphs for WebQSP/CWQ).
2. **Subgraph Retrieval:** a PCST optimization [2] extracts a compact, question-relevant subgraph that fits the LLM's context window.
3. **Graph-Constrained Decoding:** we build a KG-Trie from this subgraph and restrict a specialized LLM to it [1], so reasoning paths follow real KG connections before a general LLM deduces the final answer.

5. Baselines and Evaluation

Baselines: (i) *LLM-only* (e.g., Llama-3-8B with standard and Chain-of-Thought prompting, no KG), to measure the value of external knowledge; (ii) *KG-augmented LLMs*, namely standard RAG and RoG [4]; (iii) *full-KG GCR*, with the KG-Trie built over all L-hop paths, to test whether the PCST subgraph helps.

Metrics: *Hit@1* and *F1* for answer accuracy; the *Faithful Reasoning Ratio* (share of generated reasoning paths that exist in the KG) for faithfulness; and answer-path recall in the retrieved subgraph and latency for efficiency.

6. Expected Contribution

The project aims to produce an end-to-end KGQA pipeline that combines the scalable retrieval of GraphRAG with the KG-grounded reasoning of graph-constrained decoding. We expect it to improve the faithfulness and interpretability of LLM answers on complex questions, with visualizable reasoning paths.

7. Connection to the Course

The project combines several course topics. **Knowledge representation and Semantic Web:** Freebase and Wikidata are KGs of RDF-style triples, the latter queried with SPARQL, and reasoning paths are sequences of triples. **Language models and probability:** LLMs use autoregressive probabilities $P(x_t \mid x_{<t})$, while the KG-Trie restricts decoding to valid KG paths by masking invalid tokens. **NLP:** question answering and SentenceBERT embeddings for semantic similarity. **Human–AI interaction and explainability:** faithful, visualizable reasoning paths make answers verifiable and interpretable.

8. Checkpoints and Timeline

- **Checkpoint 1 (3 December):** data pipeline (WebQSP/CWQ and Freebase subgraph) and SPARQL graph acquisition running, baselines (LLM-only and RAG) evaluated, and PCST subgraph retrieval implemented.
- **Checkpoint 2 (7 December):** KG-Trie and graph-constrained decoding integrated into the hybrid pipeline, with a full evaluation (Hit@1, F1, faithfulness) against the baselines and a first error analysis.
- **Final report:** 12 Dec 2026; **Presentation:** 14–17 Dec.

References

[1] L. Luo et al. Graph-constrained reasoning: Faithful reasoning on knowledge graphs with large language models. In *ICML*, 2025.

[2] X. He et al. G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering. In *NeurIPS*, 2024.

[3] L. Chen et al. Plan-on-Graph: Self-correcting adaptive planning of large language model on knowledge graphs. In *NeurIPS*, 2024.

[4] L. Luo et al. Reasoning on graphs: Faithful and interpretable large language model reasoning. In *ICLR*, 2024.

Use of generative AI: generative AI tools were used to help draft and revise this proposal.